Final Project Proposal:
Digital Quantum Simulation of Physical Systems via Trotterization and Advanced Methods

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Problem Statement & Why It Matters

We will study efficient digital simulation of quantum dynamics, i.e., approximating $U(t) = e^{-iHt}$ for physically motivated Hamiltonians. Accurately simulating time evolution enables computing ground-state energies, response functions, and nonequilibrium dynamics in chemistry and condensed matter. While product-formula (Trotter–Suzuki) methods are straightforward and hardware-friendly, state-of-the-art approaches (LCU, QSP) offer superior asymptotic precision scaling. Our project targets a technically honest, end-to-end comparison anchored in *implementations* and *quantitative* evaluation on small instances.

Key Question. How do resource–accuracy trade-offs differ between Trotterization and advanced methods when applied to (i) the H_2 molecular Hamiltonian and (ii) short Heisenberg chains? What counts as success on near-term backends versus ideal simulators?

Scope & Deliverables

We will implement, analyze, and visualize digital simulation pipelines in **Qiskit**. Concretely:

1. **H_2 (static property)**: Build the second-quantized electronic Hamiltonian, map to qubits (Jordan–Wigner and Bravyi–Kitaev for comparison), and estimate the ground energy using Trotterized time evolution within Iterative Phase Estimation (IPEA). Provide error vs. Trotter-steps curves and compare to classical exact diagonalization.

2. **Heisenberg XYZ spin chain (dynamics)**: Simulate time evolution of $N = 3$–4 qubits under $H = \sum_i (J_x X_i X_{i+1} + J_y Y_i Y_{i+1} + J_z Z_i Z_{i+1} + h Z_i)$ from simple product states. Plot observables $\langle Z_i(t) \rangle$ and equal-time correlators; visualize Trotter error vs. step number.

3. **Analytic comparison to LCU/QSP**: Summarize circuits and complexity (*depth*, *gate count*, *ancilla/qubits*) using canonical references; discuss crossover regimes

where advanced methods outperform product formulas, and why near-term hardware may still favor Trotter.

Background & References Beyond Class Notes

We will draw on the following core works:

- Suzuki product formulas and error bounds ([1]).
- Truncated-Taylor/LCU Hamiltonian simulation with near-optimal scaling ([2]).
- Quantum Signal Processing (QSP) and qubitization for optimal simulation scaling ([3]).
- Qiskit Nature and operator-flow docs for chemistry workflows ([5]).
- Standard mappings (Jordan–Wigner, Bravyi–Kitaev) and small-molecule case studies ([4]).

These let us precisely define *resource scaling* (depth, total two-qubit gates, and ancilla) versus *precision scaling* (dependence on ϵ), and connect asymptotics to constant overheads.

Methods & Planned Experiments

Hamiltonian preparation. For H_2, generate integrals at a fixed bond length (e.g., $R = 0.74\,\text{Å}$) in a minimal basis (STO-3G), build the second-quantized Hamiltonian, and map to qubits via JW/BK to obtain a Pauli-sum $H = \sum_k \alpha_k P_k$. For the Heisenberg chain, construct XYZ couplings with tunable J_x, J_y, J_z, h.

Circuits.

1. **Trotterization** (first- and second-order): Implement n-step formulas $\left[\prod_k e^{-i\alpha_k P_k\, t/n}\right]^n$ and the symmetric second-order variant. Count depth and CNOTs per step.
2. **IPEA ($\mathbf{H_2}$)**: Use Trotterized controlled-$U(2^m t)$ blocks for $m = 0, \ldots, M-1$; extract the ground eigenphase and convert to energy. Benchmark against exact diagonalization.
3. **Dynamics (Heisenberg)**: Prepare basis states (e.g., $|100\cdots\rangle$), evolve to $t_{\max}$ with varying n, and measure $\langle Z_i(t)\rangle$; provide heatmaps and fidelity vs. exact evolution from matrix exponentiation.

Advanced methods (analysis). We will not fully implement LCU/QSP (time-limited), but will (i) detail circuit primitives (select/prepare oracles, phase factors), (ii) tabulate asymptotic costs $\tilde{O}(\tau\,\mathrm{polylog}(1/\epsilon))$ with $\tau = t\|H\|$, and (iii) discuss realistic constants (ancilla cost, oracle synthesis) and noise sensitivity.

Evaluation Criteria (Success Metrics)

- **Accuracy**: H_2 ground energy within < 0.01 Hartree of exact value at the chosen geometry; Heisenberg dynamics fidelity $F > 0.9$ for moderate t on noiseless simulators.
- **Scaling curves**: Plots of (error vs. Trotter steps n), (depth & CNOTs vs. n), and (runtime vs. n).
- **Clarity of comparison**: A table contrasting Trotter vs. LCU/QSP on precision scaling, resource metrics, constant factors, and hardware feasibility.
- **Reproducibility**: Clean notebooks/circuits with fixed seeds and configuration documented.

Anticipated Risks & Mitigations

Noise: On real devices, decoherence may dominate Trotter error. We will primarily use noiseless simulators and, if time permits, include one backend run with basic error mitigation (measurement error mitigation, dynamical decoupling). **Time/complexity**: Full LCU/QSP implementation is out of scope; we mitigate by providing precise resource accounting from literature and small illustrative subroutines where possible. **Resource limits**: If IPEA depth is too large, we will (i) cap M, (ii) use smaller t and tighter trotterization, or (iii) switch to phase-kickback with shorter evolutions.

Timeline & Division of Labor

Oct 30–Nov 6	Finalize literature grid; build H_2 and Heisenberg Hamiltonians; verify JW/BK mappings on tiny instances.
Nov 7–Nov 14	Implement first/second-order Trotter circuits; implement IPEA scaffolding; unit tests vs. exact exponentials.
Nov 15–Nov 22	Run sweeps over n; collect energy/dynamics data; generate fidelity and resource plots; optional noisy-backend trial.
Nov 23–Dec 5	Write analysis & crossover discussion (Trotter vs. LCU/QSP); finalize figures; prepare final report.

Roles: Eric Kim – circuit implementation, experiments, analysis. Michael Lee – literature synthesis, plotting, report assembly.

Expected Outcome

A concise, teachable comparison grounded in working circuits: end-to-end H_2 energy estimation and Heisenberg dynamics with quantitative Trotter error/resource curves, plus a principled discussion of when and why LCU/QSP win in theory and what that implies for near-term practice.

References

[1] M. Suzuki, "Fractal decomposition of exponential operators with applications to many-body theories and Monte Carlo simulations," *Phys. Lett. A* **146**(6), 319–323 (1990).

[2] D. W. Berry, A. M. Childs, and R. Kothari, "Hamiltonian simulation with nearly optimal dependence on all parameters," *Phys. Rev. Lett.* **114**, 090502 (2015).

[3] G. H. Low and I. L. Chuang, "Optimal Hamiltonian simulation by quantum signal processing," *Quantum* **1**, 33 (2017).

[4] J. T. Seeley, M. J. Richard, and P. J. Love, "The Bravyi–Kitaev transformation for quantum computation of electronic structure," *J. Chem. Phys.* **137**, 224109 (2012).

[5] Qiskit Nature Documentation: https://qiskit.org/ecosystem/nature/